

Low Back Pain

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Low back pain and [neck pain](#) are among the most common reasons for health care visits. The pain usually results from problems with the [musculoskeletal system](#)—most notably the spine, including the bones of the spine (back bones, or vertebrae), discs, and the [muscles](#) and [ligaments](#) that support it. Occasionally, low back pain results from a disorder that does not involve the musculoskeletal system.

Low back pain becomes more common as people age, affecting more than half of people over 65. It is very costly in terms of health care payments, disability payments, and missed work.

The spine (spinal column) consists of back bones (vertebrae). There are shock-absorbing discs between each of the vertebrae. The discs have a tough, outer layer of fibrocartilage and a soft, jelly-like interior called the nucleus. Each of the vertebrae have two joints behind the discs. The joints are called facet joints. The facets of one vertebral body rest on the facets of the one below it, forming a joint. The facet joints and thus the spine are stabilized by ligaments and muscles, which include the following:

- Two iliopsoas muscles, which run along both sides of the spine
- Two erector spinae muscles, which run along the length of the spine behind it
- Many short paraspinal muscles, which run between the vertebrae

The abdominal muscles, which run from the bottom of the rib cage to the pelvis, also help stabilize the spine by supporting the abdominal contents. The muscles in the buttocks also help stabilize the spine. All together, these muscles are referred to as the core muscles.

Enclosed in the spine is the [spinal cord](#). Along the length of the spinal cord, the spinal nerves emerge from the sides through spaces between the vertebrae to connect with

nerves throughout the body. The part of the spinal nerve nearest the spinal cord is called the spinal nerve root. Because of their position, spinal nerve roots can be squeezed (compressed) when the spine is injured, resulting in pain.

The lower spine (lumbar spine) connects to the spine in the upper back (thoracic spine) above and to the pelvis through the sacrum below. The sacrum is the large triangular bone at the base of the spine, the lower part of which is the tailbone. The lumbar spine is flexible to allow turning, twisting and bending, and provides strength—for standing, walking, and lifting. Thus, the lower back is involved in almost all activities of daily living. Low back pain can limit many activities and reduce the quality of life.

Spinal Column and Spinal Cord



Types of back pain

Common types of back pain include local, radiating, and referred pain.

Local pain occurs in a specific area of the lower back. It is the most common type of back pain. The cause is usually a small disc injury, joint arthritis, and muscle sprains and strains. The pain may be constant and aching or, at times, intermittent and sharp. Sudden pain may be felt when an injury is the cause. Local pain can be worsened or relieved by changes in position. The lower back may be sore when touched. Muscle spasms may occur.

Radiating pain is pain that travels from the lower back down the leg. The pain can be a dull ache or be sharp and intense. It typically involves only the side or back of the leg and may travel all the way to the foot or only to the knee. Radiating pain typically indicates compression of a nerve root caused by disorders such as a [herniated disc](#), [sciatica](#), [osteoarthritis](#), or [spinal stenosis](#). Coughing, sneezing, straining, or bending over while keeping the legs straight may trigger the pain. If there is pressure on the nerve root, the pain may be accompanied by muscle weakness in the leg, a pins-and-

needles sensation, or even loss of sensation. Rarely, people lose [bladder control](#) (urinary incontinence) or [bowel control](#) (fecal incontinence).

Referred pain is felt in a different location from the actual cause of the pain. For example, some people who have a heart attack feel pain in their left arm. Referred pain from internal organs to the lower back tends to be deep and aching, and its exact location is hard to pinpoint. Typically, movement does not worsen it, unlike pain from a musculoskeletal disorder.

Causes of Low Back Pain

Most back pain is caused by disorders of the spine and the joints, muscles, ligaments, and nerve roots around it or the discs between vertebrae. Often, no single specific cause can be identified. Any painful disorder of the spine may cause reflex tightening (spasm) of muscles around the spine. This spasm may worsen the existing pain. Stress may worsen low back pain, but how it does so is unclear.

Occasionally, back pain is due to disorders outside the spine, such as cancer, gynecologic disorders (for example, [premenstrual syndrome](#)), disorders of the kidneys (for example, [kidney stones](#)), urinary disorders (for example, [infections of the kidney](#), [bladder](#), and [prostate gland](#)), digestive tract disorders (for example, [diverticulitis](#)), and disorders of major arteries near the spine.

Common causes

Common causes of low back pain include

- Injuries to muscles and ligaments
- [Osteoarthritis](#)
- [Vertebral compression fractures](#)
- A [ruptured or herniated disc](#)
- [Lumbar spinal stenosis](#)
- [Spondylolisthesis](#)
- [Fibromyalgia](#)

Injuries may occur during routine activities (for example, lifting, exercising, moving in an unexpected way) or result from trauma, such as a fall or car crash. Often no specific injured structures are identified with imaging tests, but doctors presume that some muscles and/or ligaments have been affected.

Osteoarthritis (degenerative arthritis) causes the cartilage between the facet joints to wear away and bone spurs (osteophytes) to form. This disorder is due in part to the wear and tear of years of use. People who repetitively stress one joint or a group of joints are more likely to develop osteoarthritis in that area. The discs between the vertebrae deteriorate, and the spaces between the vertebrae narrow, increasing the pressure on the facet joints, which become inflamed (arthritis) and form bone spurs in the openings for the nerve roots. With severe degeneration and loss of disc height the osteophytes in the opening can compress spinal nerve roots. All of these changes can cause low back pain as well as stiffness.

Vertebral compression (crush) fractures (fractures of spinal vertebrae) commonly develop when bone density decreases because of osteoporosis, which typically develops as people age. Vertebrae are particularly susceptible to the effects of osteoporosis. Vertebral compression fractures (which may cause sudden, severe back pain) can be accompanied by compression of spinal nerve roots (which may cause chronic back pain). However, most fractures due to osteoporosis occur in the upper and middle back and cause upper and middle rather than low back pain.

A **ruptured or herniated disc** can cause low back pain. A disc has a tough outer layer and a soft, jelly-like interior. If a disc is repeatedly overloaded by the vertebrae above and below it (as when bending forward, particularly when lifting a heavy object), the outer layer may tear (rupture), causing pain. The interior of the disc can squeeze through the tear, so that part of the interior bulges out (herniates). This bulge can compress, irritate, and even damage the spinal nerve root next to it, causing more pain and symptoms that are felt in one or both legs. A ruptured or herniated disc in the low back that affects nerves commonly causes sciatica. However, imaging studies such as magnetic resonance imaging (MRI) often show bulging discs in people who have no symptoms or problems.

Lumbar spinal stenosis is narrowing of the spinal canal, which runs through the center of the spine and contains the spinal cord and the bundle of nerves that extends downward from the bottom of the spinal cord in the lower back. It is a common cause of low back pain in older people. Spinal stenosis also develops in middle-aged people who were born with a narrow spinal canal. Spinal stenosis is caused by such disorders as osteoarthritis and spondylolisthesis. Rarely, spinal stenosis is seen in Paget disease of bone that causes bone expansion.

Spinal stenosis may cause sciatica as well as low back pain.

Spondylolisthesis is partial displacement of a vertebra in the lower back. One type of spondylolisthesis usually occurs during adolescence or young adulthood (often in athletes) caused by an injury that fractures a part of the vertebra. If both sides of the

vertebra are involved, the vertebra can then slip forward over the one below it. Spondylolisthesis can also occur in older adults but mainly as the result of a degenerative condition. People who develop spondylolisthesis as adults are at risk of developing lumbar spinal stenosis.

Fibromyalgia is a common cause of pain that affects many parts of the body, sometimes including the low back. This disorder causes chronic widespread (diffuse) pain in muscles and other soft tissues in areas outside the lower back. Fibromyalgia is also characterized by poor sleep and fatigue.

Did You Know...

- Strengthening abdominal muscles, as well as back muscles, helps support the spine and prevent low back pain.

Less common causes

Less common causes of low back pain include

- Spinal infections
- [Spinal tumors](#)
- A bulge (aneurysm) in the large artery in the abdomen ([abdominal aortic aneurysm](#))
- Certain digestive disorders, such as [diverticulitis](#)
- Certain urinary tract disorders, such as [kidney infections](#), [kidney stones](#), and prostate infections
- Certain disorders involving the pelvis, such as [ectopic pregnancy](#), [pelvic inflammatory disease](#), and [cancer of the ovaries](#) or [other reproductive organs](#)
- [Shingles](#) (before and after the rash is seen)
- [Paget disease of bone](#)
- Several types of inflammatory arthritis, such as [ankylosing spondylitis](#)
- Inflammatory or infiltrative disorders in the space behind the abdominal cavity (retroperitoneum), such as scarring, bleeding, enlarged lymph nodes, and [immunoglobulin G4-related disease \(IgG4-RD\)](#)
- Inflammatory muscle disorders, such as [polymyositis](#) and other inflammatory myopathies and [polymyalgia rheumatica](#)

Evaluation of Low Back Pain

The doctor aims to identify any serious disorders. Because low back pain is often caused by several problems, diagnosing a single cause may not be possible. Doctors may only be able to tell that the cause is a musculoskeletal disorder and how serious it is likely to be.

Warning signs

In people with low back pain, certain symptoms and characteristics are cause for concern. They include

- A history of cancer
- Use of [medications that suppress the immune system](#), [HIV infection or late-stage HIV](#), [use of injected substances](#), recent surgery, or a wound—conditions that increase the risk of infection
- Numbness, weakness in one or both legs, difficulty emptying the bladder ([retention of urine](#)), or loss of bladder or bowel control ([urinary incontinence](#) or [fecal incontinence](#))—symptoms that suggest nerve damage or [spinal cord compression](#)
- Fever
- Weight loss
- Severe pain at night
- Abdominal or chest pain, or pulsing sensation in the upper abdomen—symptoms that suggest an [abdominal aortic aneurysm](#)
- Vomiting, severe abdominal pain, or stool that is black or bloody—symptoms that suggest a digestive disorder
- Difficulty urinating, blood in the urine, or severe crampy pain on one side radiating into the groin—symptoms that suggest a urinary tract disorder

When to see a doctor

People should see a doctor immediately if they have fever or any of the warning signs that suggest nerve damage, an [abdominal aortic aneurysm](#), a digestive disorder, or a urinary tract disorder. People with most other warning signs should see a doctor within a day. If pain is not severe and people have no warning signs other than pain for more than 6 weeks, their need to see a doctor is not as urgent.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause and the tests that may need to be done (see table

Some Causes and Features of Low Back Pain

Doctors ask questions about the pain:

- What is the pain like?
- When and how did it start?
- How severe is it?
- Where is it and where does it radiate to?
- What relieves or worsens it (for example, changes in position or weight bearing)?
- Are there other symptoms (such as numbness, weakness, retention of urine, or incontinence)?

Certain characteristics of the pain can give clues to possible causes:

- Pain in an area that is tender to the touch and is worsened by changes in position or weight bearing is usually local pain.
- Pain that radiates down the leg, such as sciatica, is usually caused by compression of a spinal nerve root.
- Pain that is not affected by changes in position of the back and is not accompanied by tenderness may be referred pain.
- Pain that is constant, severe, progressively worse, and unrelieved by rest, particularly if it keeps the person awake at night, can be a disc herniation but may indicate cancer or an infection.

The physical examination is focused on the spine and on evaluation of the nerves to the groin and legs to look for signs of nerve root compression. Signs of nerve root compression depend on which nerve roots are involved and include weakness of one of the muscle groups in a leg, abnormal reflexes (tested by tapping the tendons below the knee and behind the ankle), decreased sensation in an area of the leg, and, very rarely, retention of urine and incontinence of urine or stool (fecal incontinence).

Doctors may ask the person to move in certain ways to determine the type of pain. Doctors typically ask the person to bend forward and backward. They may have the person lie flat and then lift their leg without bending the knee to see if it brings on pain, which would suggest a herniated disc. Doctors may also check a person's abdomen for tenderness or a mass and the pulses, particularly in people over 55, who

may have an [aortic aneurysm](#). They may examine the prostate in men by doing a digital rectal examination and the internal reproductive organs in women by doing a pelvic examination.

With information about the pain, the person's medical history, and results of a physical examination, doctors may be able to determine the possible cause.



Some Causes and Features of Low Back Pain

Cause	Common Features*	Tests†
Common causes		
Sprains and strains	Pain that • Often occurs on one or both sides of the spine • Worsens with movement and lessens with rest • Typically develops while lifting, bending, or twisting	A doctor's examination
<u>Osteoarthritis</u> , sometimes with compression of a spinal nerve root	Pain over a specific part of the spine, which sometimes • Is worsened by standing • Is relieved by sitting • Travels down a leg • Is accompanied by numbness and/or weakness	X-rays Sometimes MRI or CT (for <u>osteoarthritis</u> that is severe enough to cause nerve root pain)
Usually in older people with pain and/or deformities in other joints		
<u>Vertebral compression fractures</u>	Pain over a specific part of the spine, sometimes starting suddenly	X-rays Sometimes CT or MRI



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